

Crystal Structure.

Crystal falls under category of solids

There are 2 types of solid:-

- > Amorphous solids
- > Crystalline solids

Amorphous Solids

Crystalline Solids.

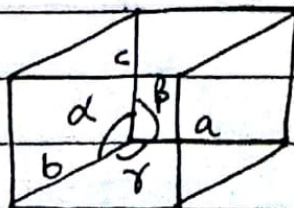
- 1) Supercondensed liquids
- 2) Periodicity remains same for a short range
- 3) They are isotropic
- 4) Do not have sharp MP & BP.
- 5) Irregular shape.
eg. Rubber, Plastics.

- 1) True solids
- 2) Periodicity remains for large range.
- 3) They are Anisotropic.
- 4) Sharp melting and Boiling point
- 5) Regular shape
eg. Quartz, Calcite

Basis: One or more than one atoms.

Lattice: Points

Crystal: Basis + Lattice.



$a = b = c$
 $\alpha = \beta = \gamma = 90^\circ$ for cubical structure

Here, $a, b, c, \alpha, \beta, \gamma$ are primitive no.

Bravais Lattice.

There are 7 categories

Cubic
Tetragonal
Orthorhombic
Monoclinic
Triclinic
Trigonal
Hexagonal

Inke further types
Ka total kinds, we
get 14 different
structures.

4 types of Basic crystals

- Primitive (P)
- Body centred (I)
- Face centred (F)
- Base centred (S)

~~4~~ 4 Values to Be calculated Now

Number of atoms per unit cell.

Coordination Number.

Atomic Radius.

Packing fraction.

Cube

Simple Cube
(S)

1) Number of Atoms :- $\frac{1}{8} \times 8 = 1$

BCC

Number of Atoms, $N = \frac{1}{8} \times 8 + 1 = 2$

FCC Number of Atoms, $N = \frac{1}{8} \times 8 + \frac{1}{2} \times 6 = 4$

Coordination Number. (Atoms nearby ~~also~~ the atom of cell)

SC: 6

BCC: 8

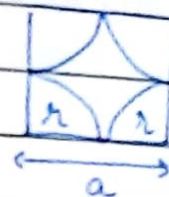
FCC:

Atomic Radius.

SC

$$a = 2r$$

$$r = a/2$$

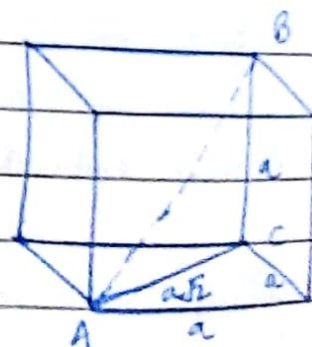


BCC

$$(a\sqrt{2})^2 + a^2 = BA^2$$

$$2a^2 + a^2 = BA^2$$

$$BA = a\sqrt{3}$$



$$2r + 2r = a\sqrt{3}$$

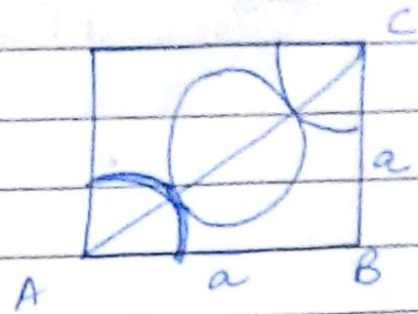
$$r = \frac{a\sqrt{3}}{4}$$

FCC

$$AC^2 = AB^2 + BC^2$$

$$16r^2 = 2a^2$$

$$r = \frac{a\sqrt{2}}{4}$$



Packing fraction:

$\frac{\text{Total volume of Atoms}}{\text{Total volume of cube}}$

SC

$$P.F = \frac{1 \times \frac{4}{3}\pi \left(\frac{a}{2}\right)^3}{a^3}$$

$$= \frac{4}{3} \pi \times \frac{1}{8} = \frac{\pi}{6}$$

BCC

$$P.F = \frac{2 \times \frac{4}{3}\pi \left(\frac{a\sqrt{3}}{4}\right)^3}{a^3}$$

$$= \frac{2 \times 4\pi \times \frac{3\sqrt{3}}{4^3}}{3}$$

$$= \frac{\sqrt{3}\pi}{8}$$

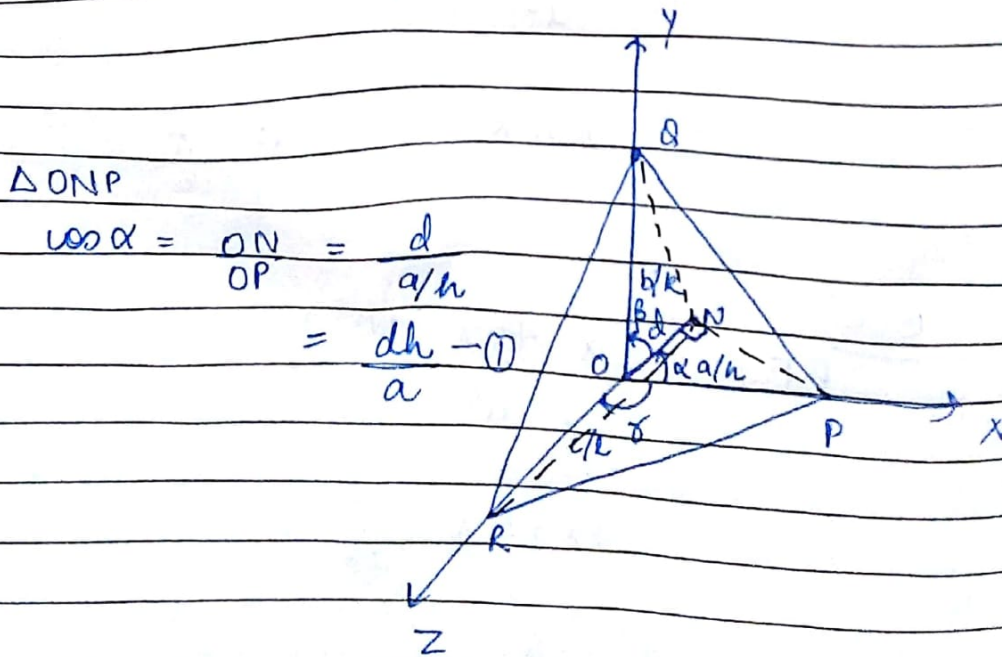
$$PF = 4 \times \frac{4\pi}{3} \left(\frac{\sqrt{2}a}{4} \right)^3$$

$$= \frac{\pi}{3} \times \frac{2\sqrt{2}}{4}$$

$$= \frac{\sqrt{2} \pi}{6}$$

→ plane (h, k, l)

→ Intercepts $\rightarrow \frac{a}{h}, \frac{b}{k}, \frac{c}{l}$



$$\Delta ONB, \quad \cos \beta = \frac{ON}{OB} = \frac{d}{b/k} = \frac{dk}{b} \quad \text{--- (2)}$$

$$\Delta ONR, \quad \cos \gamma = \frac{ON}{OR} = \frac{d}{c/l} = \frac{dl}{c} \quad \text{--- (3)}$$

from the law direction of cosine,

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\frac{d^2 h^2}{a^2} + \frac{d^2 k^2}{b^2} + \frac{d^2 l^2}{c^2} = 1$$

$$d^2 \left[\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2} \right] = 1$$

$$d = \frac{1}{\sqrt{\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}}}$$

$\Rightarrow$ for a cube, $a = b = c$

$$\therefore d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

Here h, k, l are mirror indices.

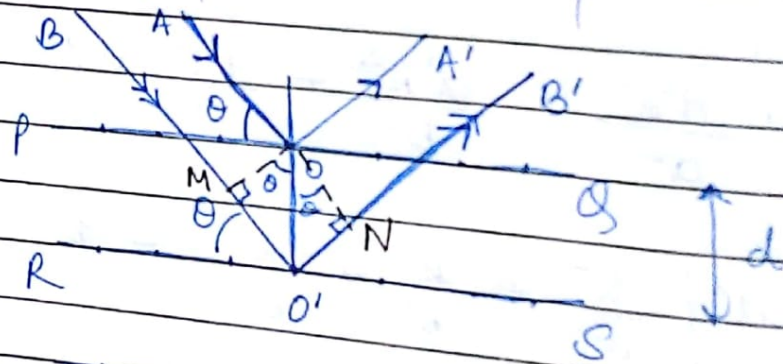
Bragg's Law.

Atoms in a crystal are arranged in a regular and periodic manner, and the spacing between them is comparable with the wavelength of X-ray.

So, a crystal can be used as a natural grating for X-ray.

$d \rightarrow$ lattice spacing.

Consider a set of parallel lattice plane.



$\theta \rightarrow$ glancing angle.

$$\text{Path dif: } \Delta = MO' + O'N$$

The path difference between the beams reflected from two planes PQ and RS = $MO' + O'N$

In $\triangle OM'O'$,

$$\sin \theta = \frac{MO'}{OO'} = \frac{MO'}{d}$$

$$MO' = d \sin \theta$$

Also, $ON' = d \sin \theta$

∴ Path difference, $\Delta = d \sin \theta + d \sin \theta$
 $= 2d \sin \theta$

$$\Delta = 2d \sin \theta$$

For the constructive interference, we know that,
this path difference must be equal to $n\lambda$.
i.e., $\Delta = n\lambda$.

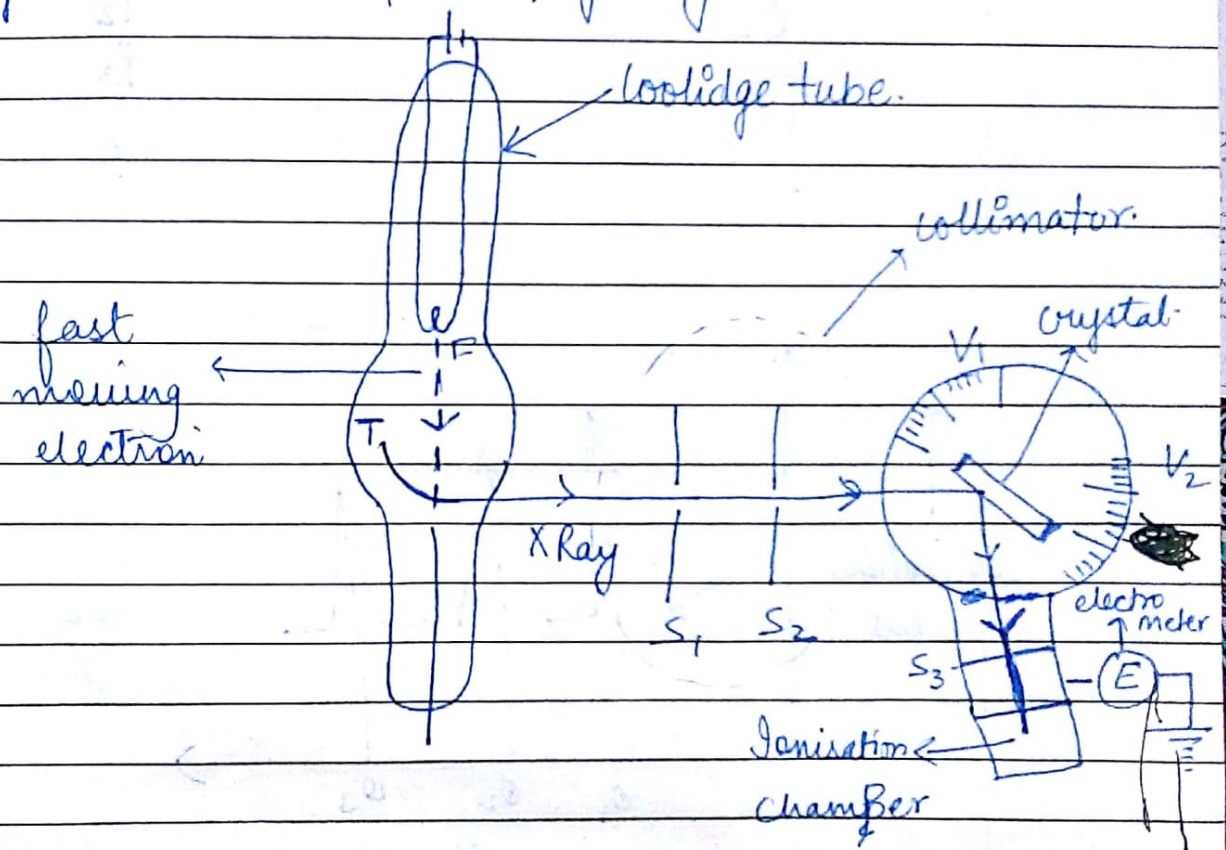
So,

$$2d \sin \theta = n\lambda$$

↓
Bragg's Equation.

Bragg's X-Ray Spectrometer

Source is X-Ray.
Crystal is used in place of grating.



coolidge tube: It produces X rays using e^- .

$$\frac{1}{2}mv^2 = eV$$

$S_1, S_2 \rightarrow$ 2 parallel slits.

In normal, grating $\rightarrow$ transmission grating
 In X Ray, crystal $\rightarrow$ acts as reflection grating.

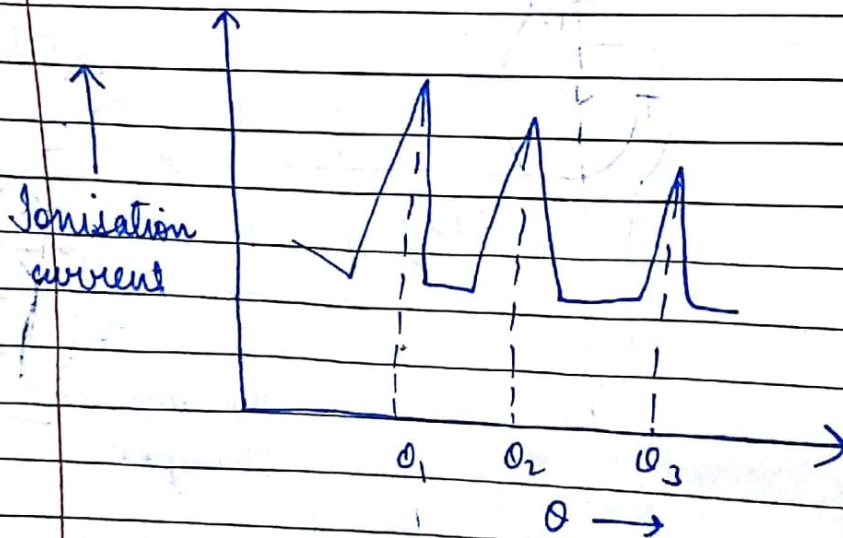
Crystal is also called Space Grating.

θ (glancing angle)

Ionisation current

θ_1
 θ_2
 θ_3
 $\vdots$

i_1
 i_2
 i_3
 $\vdots$



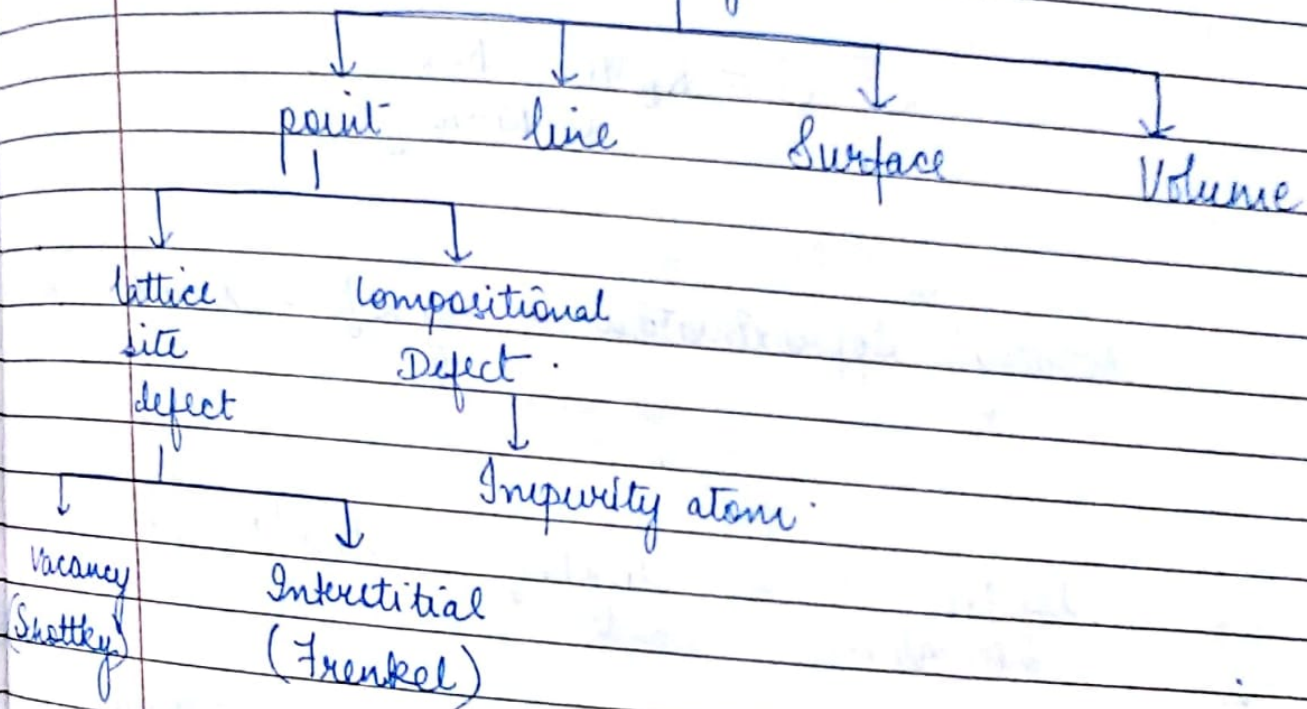
$$2d \sin \theta_1 = 1 \times \lambda$$

$$2d \sin \theta_2 = 2 \times \lambda$$

$$2d \sin \theta_3 = 3 \times \lambda$$

Imperfections In Crystal.

Imperfections in crystal.



Schottky Defect

$N \rightarrow$ Total no. of atoms present

$n \rightarrow$ No. of vacancies.

$E \rightarrow$ energy required to move 1 atom from its lattice site.

$nE \rightarrow$ Total energy to remove n atoms.

$P \rightarrow$ No. of ways to select ' n ' from ' N '.

$$\therefore P = \frac{N!}{(N-n)!n!}$$

Boltzmann Entropy formula, $S = k_B W$

$$= k_B \log P$$

$$S = k_B \log \frac{N!}{(N-n)! n!}$$

Sterling's Approximation $\rightarrow \log x! = x \log x - x$

$$\log \frac{N!}{(N-n)! n!} = \log N! - \log((N-n)! n!)$$

$$= \log N! - \log(N-n)! - \log n!$$

$$= N \log N - N - (N-n) \log(N-n) + (N-n) - n \log n + n$$

$$N \log N - N - (N-n) \log(N-n) + N - n - n \log n + n$$
$$= N \log N - (N-n) \log(N-n) - n \log n$$

$$S = k_B [N \log N - (N-n) \log (N-n) - n \log n]$$

Gibbs function:

$$G = H - TS$$

↓
enthalpy

$$U + PV \quad \left[\begin{array}{l} \text{Internal} \\ \text{Energy} \end{array} + \text{Press} \times \text{Vol} \right]$$

$$G = U + PV - TS.$$

Helmholtz free energy, $F = U - TS$

$$\therefore U = nE$$

$U \rightarrow$ Internal Energy.

$$\begin{aligned} \text{So, } F &= nE - TS \\ &= nE - T k_B [N \log N - (N-n) \log (N-n) - n \log n] \end{aligned}$$

$$\left(\frac{dF}{dn} \right)_T = E - T k_B [0 + \cancel{n} \log (N-n) + 1 - \log n - 1]$$

$$\left(\frac{dF}{dn} \right)_T = E - T k_B \frac{\log (N-n)}{n}$$

$$\min_p \left(\frac{dF}{dn} \right) = 0$$

$$\therefore E - T k_B \log \left(\frac{N-n}{n} \right) = 0$$

$$\frac{E}{k_B T} = \log \left(\frac{N-n}{n} \right)$$

$$e^{E/k_B T} = \frac{N-n}{n}$$

$$\frac{n}{N-n} = e^{-E/k_B T}$$

$$n = (N-n) e^{-E/k_B T}$$

$$n \ll N, \quad (N-n) \propto N$$

$$n = N e^{-E/k_B T}$$

$$\text{when } n = N e^{-E/2k_B T}$$

It is for ionic crystal.
(Agar the vacancy hai, to defect)
-ne vacancy hai logi